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Series



MA/IVMA/611

2026

Paper-II – Automobile Engineering
(English Version)

Question Booklet Sl. No.

1017029

Time : 150 Minutes

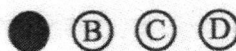
Max. Marks : 300

INSTRUCTIONS

1. The Question Paper contains **150** questions. Each question carries **2** marks. For each wrong answer $1/3^{\text{rd}}$ mark will be deducted.
2. Please check the Question Paper and ensure that it contains all the questions. If you find any defect in the Question Paper or OMR Answer Sheet, please get it replaced immediately. The OMR Answer Sheet is in Duplicate. Do not separate the copies. The markings on the first page will automatically come on the second copy also. The first copy (Original) will be retained for evaluation and the second copy (Duplicate) can be taken by the candidate.
3. Write your Registered Number on the top of the Question Paper. On the right hand corner of the first page, the Question Paper Series is printed as (A) or (B) or (C) or (D). On the OMR Answer Sheet, Side-1 at the right top corner at Part C, the Question Paper Series are printed as (A) (B) (C) (D). **Darken the appropriate circle, as per your Question Paper Series with Blue/Black Ball point pen. Gel Pens/Pencils are not allowed.**

Example to fill up the Question Paper Series on the OMR Answer Sheet.

If your Question Paper Series is A, please fill as shown below :



If you have not marked the Question Paper Series on the OMR Answer Sheet, Side-1 at Part C with Blue/Black Ball point pen, or marked in a way that it leads to discrepancy in determining the exact Question Paper Series, then your Answer Sheet will be invalidated without any further notice. No correspondence will be entertained in the matter.

4. Please get the signature of the Invigilator affixed in the space provided at the top of the OMR Answer Sheet. The OMR Answer Sheet without the signature of the Invigilator will be invalidated.



5. Each question is followed by 4 answer choices. Of these, you have to carefully select one correct answer and mark it on the OMR Answer Sheet by completely darkening the appropriate circle for the question. If more than one circle is darkened, the answer will not be valued at all. Use Blue or Black Ball point pen to make thick mark to fill the circle completely. Make **no** other stains or marks. Using **Whitener/Blade/Eraser/Chalk Powder** anywhere on the OMR Answer Sheet or any kind of tampering to change the answers on OMR Answer Sheet will lead to **invalidation** of the entire OMR Answer Sheet.

e.g. : If the answer for Question No. 1 is Answer choice (2), it should be marked as follows :

1	①	●	③	④
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6. Do **not** fold, tear, wrinkle, tie, staple, do any rough work or make any stray marks on the OMR Answer Sheet. Otherwise your OMR Answer Sheet will be **invalidated**.
7. Do **not** mark answer choices on the Question Paper. Violation of this will be viewed seriously.
8. Rough work may be done on the last page of the Question Paper only. It should not be done anywhere else.
9. **Scientific Calculator is allowed**. However, any other type of Electronic Devices such as Mathematical/Log Tables, Mobile Phones, Bluetooth instruments, Digital Watches, Programmable Calculators etc., should not be brought into the Examination Hall. If found in possession of such devices, it will be considered as cheating and strict action taken.
10. As per G.O. Ms. No. 385, GA (Ser.A) Dept., dt: 18/10/2016, if any candidate attempts to use any means to cheat/copy etc., he/she is liable for prosecution (Police Case) and debarment.
11. Candidates should stay in the Examination Hall till expiry of Full Time i.e. 2½ Hours. No candidate shall be allowed to leave the Hall in the middle of the examination under any circumstances.
12. Before leaving the examination hall, the candidate should hand over the OMR Answer Sheet top sheet (Original) to the Invigilator and carry the bottom sheet (Duplicate) for his/her record. Carrying away Original OMR Answer Sheet will be a criminal offence.



1. A system is in thermodynamic equilibrium when
- (1) It exchanges mass but not energy with its surroundings
 - (2) It exchanges both mass and energy with its surroundings
 - (3) Macroscopic variables like pressure, volume and temperature do not change with time
 - (4) It exchanges energy but not mass with its surroundings

2. The Zeroth Law of Thermodynamics states that if two systems are each in thermal equilibrium with a third system, then



- (1) They are in mechanical equilibrium
- (2) They are in chemical equilibrium
- (3) They are in thermal equilibrium with each other
- (4) They are in a state of total thermodynamic equilibrium

3. Which of the following is not an example of a pure substance ?

- (1) Water (H_2O)
- (2) Nitrogen (N_2)
- (3) Air (a mixture of gases like O_2 , N_2 , Ar)
- (4) Helium (He)



4. The Mollier diagram is a plot of

- (1) Temperature and entropy
- (2) Temperature and enthalpy
- (3) Pressure and enthalpy
- (4) Enthalpy and entropy

5. For a fixed amount of heat supplied and a fixed compression ratio, the thermal efficiency of the air-standard cycle follows the order



- (1) Otto > Diesel > Dual
- (2) Otto > Dual > Diesel
- (3) Diesel > Otto > Dual
- (4) Dual > Otto > Diesel

6. The Stirling cycle consists of

- (1) Two isothermal and two constant pressure processes
- (2) Two isothermal and two constant volume processes
- (3) Two isothermal and two adiabatic processes
- (4) Two adiabatic processes and one constant volume processes



7. The diagram factor is defined as the ratio of
- (1) Brake power to indicated power
 - (2) Indicated work done to brake work done
 - (3) Area of the actual indicator diagram to the area of the theoretical indicator diagram
 - (4) Thermal efficiency to mechanical efficiency

8. The radial distance from the pitch circle to the root of the tooth is known as the
- (1) Addendum
 - (2) Clearance
 - (3) Dedendum
 - (4) Whole depth

9. What is the correct mathematical formula for the module of a gear ?

- (1) D/N
- (2) $D/(2N)$
- (3) $2D/N$
- (4) $3D/N$

(Where D is the pitch circle diameter and N is the number of teeth)

10. In a simple gear train, how is the driven gear's motion related to the driver gear if there is an odd number of gears ?

- (1) Opposite direction
- (2) Same direction
- (3) Does not depends on gear ratio
- (4) Depends on gear ratio

11. In an epicyclic gear train, what is the purpose of the arm (or carrier) ?

- (1) It is a fixed, stationary gear
- (2) It carries the planet gears and rotates
- (3) It is the primary output shaft
- (4) It fixes the sun gear's position

12. What is the primary factor that distinguishes the torsional shear stress distribution in a solid shaft compared to a hollow shaft of the same outer diameter, when subjected to the same torque ?

- (1) Shear stress is zero at the center of both shafts
- (2) Shear stress is maximum at the center of the hollow shaft but zero at the center of the solid shaft
- (3) Shear stress is zero at the inner surface of the hollow shaft but uniform across the solid shaft
- (4) Shear stress is highest at the outer surface of the hollow shaft, while the solid shaft's stress increases linearly from the center to the outer surface



13. For a solid shaft under pure torsion, how does the angle of twist (θ) change if the shaft's length (L) is doubled, while the polar moment of inertia (J) and modulus of rigidity (G) remain constant?

- (1) It remains the same
- (2) It is halved
- (3) It doubles
- (4) It is quadrupled

14. When a shaft experiences both a bending moment (M) and torque (T) simultaneously, what is the nature of the stresses induced?



- (1) Only shear stress due to torque is present
- (2) Only normal stress due to bending is present
- (3) Only shear stress due to bending is present
- (4) Both normal stress (from bending) and shear stress (from torque) are developed in the shaft

15. A body slides down a 45° inclined plane in twice the time it would take on a frictionless surface. What is the coefficient of kinetic friction, μ_k ?

- (1) 0.33
- (2) 0.75
- (3) 0.5
- (4) 1.0

16. A screw is considered self-locking if

- (1) The friction angle is less than the lead angle
- (2) The friction angle is greater than the lead angle
- (3) The friction angle is equal to the lead angle
- (4) The friction angle is half the lead angle

17. What is the primary function of a screw jack?



- (1) To convert linear motion into rotary motion
- (2) To convert rotary motion into linear motion to lift or lower heavy loads
- (3) To measure the pitch of a screw
- (4) To create a locking mechanism for a nut



18. Which statement correctly compares flash point and fire point?

- (1) The fire point is always lower than the flash point.
- (2) The flash point and fire point are the same temperature.
- (3) The fire point is typically higher than the flash point.
- (4) The flash point is higher than fire point.



19. Which of the following fuels has the highest calorific value ?
(1) Wood (2) Coal
(3) LPG (4) Petrol
20. A device used for measuring heat exchange is called a
(1) Thermometer (2) Joule meter
(3) Calorimeter (4) Pressure gauge
21. Which of the following is the chemical that is used as the reference standard for a Cetane number of 100 ?
(1) Heptane (2) Octane
(3) Hexadecane (Cetane) (4) Naphthalene
22. Which of the following is an advantage of a multi-cylinder engine over a single-cylinder engine of the same total displacement ?
(1) Higher low-end torque
(2) Simpler construction and lower manufacturing costs
(3) Smoother operation with less vibration
(4) Better fuel efficiency at low power demands
23. The primary function of a pre-whirl chamber is to
(1) Increase the engine's compression ratio
(2) Create a high-velocity, swirling motion to improve fuel-air mixing
(3) Cool the incoming air charge
(4) Allow for the use of a high-pressure common-rail fuel system
24. Which of the following factors increases the ignition delay period in a CI engine ?
(1) Increased compression ratio (2) Increased air turbulence
(3) Supercharging (4) Lower cetane number fuel
25. Which of the following factors can increase the tendency of knocking in a Compression Ignition (CI) engine ?
(1) A longer ignition delay period
(2) Using a fuel with a higher cetane number
(3) A lower compression ratio
(4) A lower temperature of the unburnt fuel-air mixture



26. What is the typical firing order for a four-cylinder inline engine ?

(1) 1-2-3-4

(2) 1-3-2-4

(3) 1-3-4-2

(4) 4-3-2-1

27. Why is the firing order of an engine so important ?

(1) It determines the engine's horsepower rating

(2) It ensures the engine's balance, efficiency and performance by distributing power strokes evenly

(3) It is a purely aesthetic choice in engine design

(4) It affects the fuel type the engine can use

28. Which of the following is not a function of piston rings ?

(1) To seal high-pressure gases

(2) To help in detonation

(3) To maintain sufficient lubrication

(4) To transfer heat to the cylinder wall

29. Piston rings are commonly made from which material ?

(1) Aluminum alloy

(2) Brass

(3) Cast iron or alloy cast iron

(4) Steel

30. During normal operation of an internal combustion engine, where is the highest temperature typically found on the piston ?

(1) Piston pin boss

(2) Skirt of the piston

(3) Top ring groove

(4) Center of the piston crown

31. Which type of injection is considered the most accurate ?

(1) Throttle body injection

(2) Manifold injection

(3) Direct injection

(4) Port injection

32. What is the purpose of the venturi in a carburetor ?

(1) To increase the fuel flow

(2) To decrease the pressure and draw fuel into the airstream

(3) To regulate the amount of air entering the engine

(4) To provide a consistent air-fuel ratio



33. What is the purpose of internal baffles inside a fuel tank ?

- (1) To increase the tank's volume
- (2) To reduce fuel sloshing during movement
- (3) To allow for easier fuel delivery
- (4) To help the fuel pump run cooler

34. The unbalanced mass in a single-cylinder engine is balanced by counteracting the centrifugal force of the disturbing mass with the centrifugal force of a balancing mass. This is done for which of the following ?

- (1) Dynamic balancing
- (2) Neither static nor dynamic balancing
- (3) Both static and dynamic balancing
- (4) Static balancing

35. The primary unbalanced force in a single-cylinder engine is maximum when the crank angle (θ) is

- (1) 0° and 90°
- (2) 0° and 180°
- (3) 90° and 180°
- (4) 180° and 360°

36. What is the purpose of a balancing mass in a single-cylinder engine ?

- (1) To increase the speed of the engine
- (2) To balance the rotating parts and reduce vibration
- (3) To provide torque to the engine
- (4) To reduce the engine's stroke

37. Which type of lubrication system is most commonly used in two-stroke engines ?

- (1) Splash lubrication system
- (2) Wet sump lubrication system
- (3) Pressure lubrication system
- (4) Mist lubrication system

38. Which of the following is a component of a pressure lubrication system ?

- (1) Scoop on the connecting rod
- (2) Oil mixed with fuel
- (3) Oil pump that forces oil through channels
- (4) Oil splashed onto components from a reservoir



39. Which of the following is a key advantage of the liquid cooling system ?
- (1) It is very cheap
 - (2) It provides even cooling throughout the engine
 - (3) It is dependent only on the supply of water
 - (4) It absorbs a considerable amount of power from the engine
40. In an air-cooled engine, what component helps to increase the surface area for heat dissipation ?
- (1) Radiator fan
 - (2) Pressure cap
 - (3) Extended fins
 - (4) Circulating air
41. What is the role of the dog clutch in a synchromesh gearbox after synchronization ?
- (1) To initiate the speed matching process
 - (2) To provide the final positive engagement between the gear and the shaft
 - (3) To disengage the gears during a shift
 - (4) To absorb vibrations during operation
42. What is the primary function of a torque converter in an automatic transmission ?
- (1) To engage and disengage the engine from the transmission
 - (2) To transmit power from the engine to the transmission
 - (3) To provide a fluid coupling to transmit torque from the engine to the transmission
 - (4) To control the gear selection and shift points
43. Which of the following is a basic component of most automatic transmissions ?
- (1) Flywheel
 - (2) Planetary gear set
 - (3) Friction clutch
 - (4) Clutch pedal
44. What is the purpose of a slip joint in a propeller shaft ?
- (1) To allow the shaft to bend sideways
 - (2) To transfer torque at an angle
 - (3) To transmit torque when the shafts are not parallel
 - (4) To allow the shaft to change length due to expansion or contraction
45. Which type of joint is used to allow the propeller shaft to transmit torque at an angle ?
- (1) Universal joint
 - (2) Hinge joint
 - (3) Ball and socket joint
 - (4) Pivot joint



46. A differential allows the rear wheels to rotate at different speeds. This is essential for

- (1) Starting the vehicle on a hill
- (2) Increasing the vehicle's horsepower
- (3) Proper steering during a turn
- (4) Applying the brakes

47. Which type of clutch does not require a clutch pedal ?

- (1) Single plate
- (2) Multi-plate
- (3) Centrifugal
- (4) Cone

48. In a single plate clutch, what is the function of the pressure plate ?

- (1) To apply friction to the clutch disc
- (2) To disengage the clutch disc from the flywheel when the pedal is depressed
- (3) To hold the clutch disc between the flywheel and itself when the pedal is released
- (4) To absorb heat generated during engagement

49. Why are multi-plate clutches used instead of single plate clutches ?

- (1) To reduce torque transmission capacity
- (2) To increase the torque transmission capacity
- (3) To operate as a wet clutch only
- (4) To increase the overall size of the clutch

50. In a disc brake, which component is responsible for applying pressure to the pads ?

- (1) Brake drum
- (2) Piston in the caliper
- (3) Brake fluid
- (4) Bleed screw

51. What is the main advantage of disc brakes over drum brakes ?

- (1) They are generally less expensive
- (2) They are less prone to brake fade due to better heat dissipation
- (3) They are simpler in design and require less maintenance
- (4) They are more effective when the vehicle is in reverse gear

52. Which of these is an advantage of hydraulic brakes over mechanical brakes ?

- (1) They require more pedal effort for the same braking effect
- (2) They require less pedal effort for the same braking effect
- (3) They only provide uniform braking force to one wheel
- (4) They are less effective at high speeds



53. Which vehicle type typically uses air brakes due to their size and weight ?

- (1) Passenger cars
- (2) Motorcycles
- (3) Trucks, buses and semi-trailers
- (4) Electric scooters

54. When a vehicle turns, the inner front wheel must

- (1) Turn at a smaller angle than the outer wheel
- (2) Turn at the same angle as the outer wheel
- (3) Turn at a greater angle than the outer wheel
- (4) Have a negative steering angle

55. The condition for correct steering is that all four wheels of a vehicle must

- (1) Turn at the same angle
- (2) Turn about the same instantaneous center
- (3) Have the same turning radius
- (4) Be perfectly aligned at all times



56. What is the primary function of the Ackermann steering mechanism ?

- (1) To make all four wheels turn at the same angle
- (2) To achieve true rolling by ensuring the wheels turn at different angles when cornering
- (3) To provide power steering assist to the driver
- (4) To improve fuel efficiency

57. What is the function of the alternator in a vehicle ?

- (1) To provide the initial power to start the engine
- (2) To recharge the battery and power the electrical accessories
- (3) To convert electrical energy to mechanical energy
- (4) To regulate the oil pressure

58. Which component converts mechanical energy into electrical energy to charge the battery and power the electrical system ?

- (1) Starter motor
- (2) Alternator
- (3) Regulator
- (4) Ignition coil





59. What is the purpose of a relay in an automotive electrical circuit ?
- (1) To reduce the voltage in the circuit
 - (2) To increase the resistance of the circuit
 - (3) To act as a remote-controlled switch, allowing a low-current circuit to control a high-current circuit
 - (4) To store electrical charge for later use
60. What is the impact of high concentrations of pollutants like sulfur dioxide on the environment ?
- (1) Causes a cooling effect in the atmosphere
 - (2) Is the primary cause of acid rain
 - (3) Do not cause the formation of smog
 - (4) Neutralizes acid in water bodies
61. What is a major exhaust emission from CI (diesel) engines compared to SI (petrol) engines ?
- (1) Carbon Monoxide (CO)
 - (2) Particulates
 - (3) Sulphur
 - (4) Unburnt Hydrocarbons (HC)
62. Which of the following factors increases the tendency for NO_x formation in an SI engine ?
- (1) Decreasing the air-fuel ratio
 - (2) Increasing the peak combustion temperature
 - (3) Increasing the ignition delay
 - (4) Lowering the engine's compression ratio
63. Odor emissions from diesel engines are mainly caused by
- (1) Oxides of nitrogen (NO_x)
 - (2) Carbon monoxide (CO)
 - (3) Particulate matter (soot)
 - (4) Unburnt or partially burnt hydrocarbons, including aldehydes
64. Aldehydes in diesel engine exhaust are a type of
- (1) Sulfur oxide compound
 - (2) Particulate matter
 - (3) Unburnt hydrocarbon (HC)
 - (4) Inert gas
65. Which of the following operating conditions is a primary cause for high NO_x emissions in a diesel engine ?
- (1) Low combustion temperature
 - (2) Insufficient oxygen supply
 - (3) High combustion pressure and temperature with sufficient reaction time
 - (4) Delayed injection timing



66. Diesel engine exhaust primarily contains which forms of nitrogen oxides ?

- (1) N_2O and NO_2
- (2) N_2O_4 and N_2O_3
- (3) Pure N_2 and O_2
- (4) NO and NO_2

67. What is the function of the condenser in an automotive air-conditioning system ?

- (1) It cools the refrigerant
- (2) It compresses the refrigerant
- (3) It expands the refrigerant
- (4) It removes moisture from the refrigerant

68. What is the purpose of the recirculating mode in a car's air-conditioning ?

- (1) To pull in outside air
- (2) To increase the use of the AC system
- (3) To recycle air already inside the cabin
- (4) To turn the AC system off

69. The outward or inward tilt of a wheel from the vertical is known as

- (1) Caster
- (2) Toe
- (3) Camber
- (4) Kingpin inclination

70. Which of the following steering geometry components is the lateral inward tilt of the steering axis from the vertical ?

- (1) Caster angle
- (2) Camber
- (3) Kingpin inclination
- (4) Toe-in

71. A vehicle set with "toe-out" on the front wheels will typically exhibit which handling characteristic ?

- (1) Increased straight-line stability at high speeds
- (2) Slower and smoother steering response
- (3) Enhanced steering sensitivity and quicker turn-in to corners
- (4) Reduced tyre wear over time

72. Where will an underinflated tyre wear the tread most ?

- (1) Near the center
- (2) Near the edge (both edges)
- (3) In the cross direction
- (4) Evenly across the surface

73. Grooves (tread) are made in the tyres of vehicles to

- (1) Allow the tyre to fit the rim better
- (2) Make the tyre look more appealing
- (3) Improve traction and grip on the road, especially in wet conditions
- (4) Reduce the weight of the tyre

74. Which of the following is a common crystal structure for metals ?

- (1) Simple Cubic (2) Body-Centered Cubic (BCC)
(3) Octagonal (4) Triangular

75. Which of the following non-ferrous metals is known for its high electrical and thermal conductivity ?

- (1) Lead (2) Zinc
(3) Aluminum (4) Tin

76. Which element is the main component in ferrous metals ?

- (1) Zinc (2) Tin
(3) Copper (4) Iron

77. The ability of a material to be drawn into wires is called

- (1) Malleability (2) Ductility
(3) Toughness (4) Hardness

78. A thermocouple is a type of transducer used to measure

- (1) Force (2) Pressure
(3) Displacement (4) Temperature

79. An automatic cruise control system in a car is an example of a

- (1) Open-loop system (2) Closed-loop system
(3) Simple control system (4) Feedforward control system

80. Which of the following is commonly used in automobiles and other applications to measure force or weight by converting it into an electrical signal ?

- (1) Thermistor (2) Load cell (Force Transducer)
(3) Thermocouple (4) Potentiometer

81. What is the highest possible theoretical efficiency of a heat engine operating with a hot reservoir of furnace gases at 2100°C when the cooling water available is at 15°C ?

- (1) 0.678 or 67.8% (2) 0.778 or 77.8%
(3) 0.878 or 87.8% (4) 0.978 or 97.8%

82. The specific volume of water when heated at 0°C

- (1) First increases and then decreases (2) First decreases and then increases
(3) Increases steadily (4) Decreases steadily



83. What is the value of absolute zero scale of temperature ?

- (1) 273 °C
- (3) - 273 °K

(2) 273 °K

(4) 0° K

84. The efficiency of an Otto Cycle is 60% and $\gamma = 1.5$. What is the compression ratio ?

(1) 4.25

(2) 5.25

(3) 6.25

(4) 7.25

85. Mean effective pressure of the Otto Cycle is

(1) Inversely proportional to pressure ratio

(2) Directly proportional to pressure ratio

(3) Proportional to square root of pressure ratio

(4) Does not vary with pressure ratio

86. The product of diametral pitch (P) and circular pitch (p) is

(1) π

(2) 2π

(3) 1

(4) 0

87. A single reduction gear of 120 kW with a pinion 250 mm pitch circle diameter and 650 r.p.m. is supported in bearings on either side. Calculate the torque transmitted in N-m.

(1) 1765

(2) 1675

(3) 1567

(4) 1176

88. The train value of a gear train is

(1) equal to velocity ratio of a gear train

(2) reciprocal of velocity ratio of a gear train

(3) always greater than unity

(4) always less than unity

89. Flash point and fire point are characteristics primarily related to the

(1) Low-temperature behaviour of the fuel

(2) High-temperature behaviour and volatility of the fuel

(3) Purity of the fuel

(4) Viscosity of the fuel



90. The fire point of a fuel is the lowest temperature at which its vapor will continue to burn for at least how many seconds after ignition ?

- (1) 1 second
- (2) 2 seconds
- (3) 4 seconds
- (4) 5 seconds



91. A line shaft rotating at 200 r.p.m. is to transmit 20 kW. The shaft may be assumed to be made of mild steel with an allowable shear stress of 42 MPa. Determine the torque transmitted by the shaft, neglecting the bending moment on the shaft in N-m.

- (1) 80 mm
- (2) 90 mm
- (3) 70 mm
- (4) 50 mm

92. Which of the following loading is considered for the design of axles ?

- (1) Bending moment only
- (2) Twisting moment only
- (3) Combined bending moment and torsion
- (4) Combined action of bending moment, twisting moment and axial thrust

93. When a shaft is subjected to a bending moment M and a twisting moment T , then the equivalent twisting moment is equal to

- (1) $M + T$
- (2) $M^2 + T^2$
- (3) $\sqrt{M^2 + T^2}$
- (4) $\sqrt{M^2 - T^2}$

94. Screws used for power transmission should have

- (1) low efficiency
- (2) high efficiency
- (3) very fine threads
- (4) strong teeth

95. A screw jack has square threads and the lead angle of the thread is α . The screw jack will be self locking when the coefficient of friction (μ) is

- (1) $\mu > \tan \alpha$
- (2) $\mu = \sin \alpha$
- (3) $\mu = \cot \alpha$
- (4) $\mu = \operatorname{cosec} \alpha$

96. What force would be required to slide it down if the coefficient of friction between body and plane is 0.3 ?

- (1) 0 kg
- (2) 2 kg
- (3) 3 kg
- (4) 5 kg



97. In a C.I. engine squish is created
(1) towards end of compression stroke
(2) at the end of suction stroke
(3) at the beginning of suction stroke
(4) during combustion
98. Detonation in S.I. engines occurs due to
(1) Preignition of charge before the spark
(2) Sudden ignition of charge before the spark
(3) Autoignition of the charge after the spark is struck
(4) Non availability of fuel
99. The ratings of C.I. engine fuel is given by
(1) Octane Number
(2) Performance Number
(3) Cetane Number
(4) Bearing Number
100. Bomb Calorimeter is used to find the calorific value of _____ fuel.
(1) Solid
(2) Gaseous
(3) Solid and gaseous
(4) Nuclear
101. In C.I. engines knocking tendency increases with
(1) Increases in compression ratio
(2) Increasing inlet temperature of air
(3) Decrease in compression ratio
(4) Increasing the coolant temperature
102. In C.I. engines with increase in compression ratio the delay period
(1) Increases
(2) Decreases
(3) First increases and then decreases
(4) Remains unaffected
103. The top ring nearest to the piston crown is known as
(1) Compression ring
(2) Oil ring
(3) Scraper ring
(4) Groove ring
104. For reducing wear and eliminate scuffing, the piston rings are
(1) Lubricated
(2) Made of cast iron
(3) Provided with stepped groove
(4) Plated with chromium or cadmium
105. IC engine pistons are usually made of
(1) Cast Steel
(2) Forged Steel
(3) Phosphor Bronze
(4) Aluminium Alloy



106. The venturi in carburetor results in

- (1) Decrease in air velocity
- (2) Increase in air velocity
- (3) Decrease in fuel flow
- (4) Increase in manifold vacuum

107. Precise petrol injection system is

- (1) Direct injection
- (2) Sequential injection
- (3) Throttle body injection
- (4) Port injection

108. The fuel injection pressure in solid injection system is around

- (1) Less than 10 bar
- (2) 10 – 20 bar
- (3) 30 – 50 bar
- (4) 200 – 250 bar

109. _____ regulates the pressure strokes in the fuel injection pump of a diesel engine.

- (1) Pump shaft
- (2) Control rack
- (3) Lift of plunger
- (4) Needle valve

110. The primary unbalanced force is maximum when the angle of inclination of the crank with the line of stroke is

- (1) 60°
- (2) 90°
- (3) 180°
- (4) 360°

111. The partial balancing means

- (1) Balancing partially the revolving masses
- (2) Balancing partially the reciprocating masses
- (3) Best balancing of engines
- (4) All of the above

112. Detergents are oil additives used to

- (1) Reduce viscosity
- (2) Increase fire point
- (3) Prevent sludge formation
- (4) Prevent foaming

113. Friction that occurs between the layers of oil film is called

- (1) Viscous friction
- (2) Greasy friction
- (3) Dry friction
- (4) Boundary friction



114. The main purpose of thermostat in an engine cooling system is to

- (1) Allow the engine to warm quickly
- (2) Prevent the coolant from boiling
- (3) Pressurise the system
- (4) Indicate to the driver the coolant temperature

115. Two advantages of using helical gears rather than spur gears in a transmission are

- (1) Strength and cost
- (2) Strength and less end thrust
- (3) Noise level and strength
- (4) Noise level and economy

116. The adjustment for backlash in a differential is provided between

- (1) Crown wheel and sun gear
- (2) Sun gear and planet gear
- (3) Crown wheel and the drive pinion
- (4) Crown wheel and planet gear

117. Central portion of the propeller shaft is made from

- (1) steel shaft
- (2) gun metal shaft
- (3) steel tube
- (4) cast iron rod

118. The smallest gears inside the differential casing are

- (1) pinion gears
- (2) sun gears
- (3) side gears
- (4) ring gears

119. The primary function of a ball and socket joint is to

- (1) Allow rotation in one plane only
- (2) Allow linear motion only
- (3) Allow movement in almost all directions
- (4) Restrict all movements

120. Two-piece propeller shaft requires

- (1) One universal joint
- (2) A centre support bearing
- (3) The shaft to be solid
- (4) None of the above

121. A single dry plate clutch is to be designed to transmit 7.5 kW at 900 r.p.m. Find the diameter of the shaft.

- (1) 21.6 mm
- (2) 26.6 mm
- (3) 32.6 mm
- (4) 36.6 mm

122. In case of a multiple disc clutch, if n_1 are the number of discs on the driving shaft and n_2 are the number of the discs on the driven shaft, then the number of pairs of contact surfaces will be

- (1) $n_1 + n_2$ (2) $n_1 + n_2 - 1$
 (3) $n_1 + n_2 + 1$ (4) $n_1 - n_2 - 1$

123. The coefficient of friction for the clutch facing is approximately

- (1) 0.1 (2) 0.4
 (3) 0.8 (4) 1.2

124. The maximum value of axial force at the clutch which a driver can apply while driving without getting fatigued is approximately

- (1) 10N (2) 100N
 (3) 500N (4) 5000N

125. In drum type brakes the fluid on releasing return to the master cylinder due to the action of the piston returns spring and

- (1) bypass port (2) wheel cylinder spring
 (3) compensating port (4) brake shoe retractor spring

126. The hand brake usually operates on

- (1) Rear wheels (2) Front wheels
 (3) Right side wheels (4) Left side wheels

127. What is the efficiency of hydraulic braking system ?

- (1) 20% – 30% (2) 40% – 50%
 (3) 65% – 75% (4) About 90%

128. _____ is not a part of hydraulic braking system.

- (1) Wheel Cylinder (2) Master Cylinder
 (3) Steering Mechanism (4) Brake Pedal

129. The brake employed in cars are usually operated

- (1) mechanically (2) hydraulically
 (3) by means of engine vacuum (4) by compressed air



130. Fading of brake occurs

- (1) at high speed
- (2) at low speed
- (3) during continuous application
- (4) when brake lining

131. The brake bleeding process removes from the system

- (1) Air
- (2) Vacuum
- (3) Excess fluid
- (4) Excess pressure

132. The steering ratio for manual steering of cars is approximately

- (1) 5
- (2) 15
- (3) 50
- (4) 100

133. The turning circle for a car is approximately

- (1) 1 metre
- (2) 2 metres
- (3) 10 metres
- (4) 30 metres

134. A vehicle has a wheelbase (L) of 2 meters and a wheel track (W, distance between front wheel pivots) of 1.5 meters. Calculate the required turning angle for the outer front wheel (ϕ) if the inner front wheel is steered to an angle (θ) of 30 degrees.

- (1) Approximately 25 degrees
- (2) Approximately 22 degrees
- (3) Approximately 20 degrees
- (4) Approximately 28 degrees

135. Three components of primary ignition circuit are

- (1) Contact breaker, condenser and distributor cap
- (2) Contact breaker, ignition coil and spark plugs
- (3) Contact breaker, ignition switch and condenser
- (4) Contact breaker, ignition switch and rotor

136. The contact breaker points are opened by the cam and closed by the

- (1) Same cam
- (2) Centrifugal force
- (3) Magnetic force
- (4) Spring tension

137. The two quantities used for specifying electrical wire are

- (1) The colour and length
- (2) The length and diameter
- (3) The diameter and number of strands
- (4) The colour and diameter

138. The frequency of a vibrating contact type for d.c. generator is

- (1) 20 per second
- (2) 200 per second
- (3) 20 per minute
- (4) 200 per minute

139. The carbon accumulation in an engine cylinder results in

- (1) Increase of effective compression ratio
- (2) Increase of volumetric efficiency
- (3) Increase of clearance volume
- (4) Increase of ignition time

140. Due to which of the following reasons the piston rings are plated with chromium, cadmium or phosphate?

- (1) To prevent clogging
- (2) To improve heat transfer
- (3) To reduce wear and eliminate scuffing
- (4) To improve surface finish

141. Carbon Monoxide emission from S.I. engine are higher when the engine is

- (1) Idling
- (2) Cruising
- (3) Accelerating
- (4) Decelerating

142. Orsat flue gas analysis gives

- (1) Mass analysis of wet products of combustion
- (2) Volumetric analysis of dry products of combustion
- (3) Mass analysis of dry products of combustion
- (4) Volumetric analysis of wet products of combustion

143. Flame Ionisation Detector is used for measuring

- (1) CO
- (2) HC
- (3) NO_x
- (4) CO₂

144. Chemiluminescence technique is used to measure

- (1) NO_x
- (2) CO
- (3) CO₂
- (4) Smoke intensity



145. Non-dispersive infra-red analyser is widely accepted instrument for measuring

- (1) NO_x
- (2) HC
- (3) CO
- (4) CO_2

146. What is depth of hardened layer by flame hardening of steel wheel ?

- (1) 0.15 mm
- (2) 0.8 mm
- (3) 0.35 mm
- (4) 1.2 mm

147. By what process maximum hardness is obtained for a steel part ?

- (1) Carburizing
- (2) Nitriding
- (3) Cyaniding
- (4) Same hardness in all processes

148. The ratio of E of a whisker of mild steel to E of a bulk material is

- (1) 4
- (2) 5
- (3) 40
- (4) 50

149. Which of the following is a possible reason for flushing an A/C condenser ?

- (1) To replace the old refrigerant with new refrigerant
- (2) To remove sludge and debris
- (3) To remove moisture
- (4) To improve cooling efficiency

150. The disadvantage of radial ply tyre compared to cross ply tyre is

- (1) uncomfortable ride at low speeds
- (2) uncomfortable ride at high speeds
- (3) higher cornering power
- (4) lower rolling resistance



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145. Non-dispersive infrared is used for measuring **SPACE FOR ROUGH WORK**

- (1) NO₂
- (2) HC
- (3) CO
- (4) CO₂



146. What is depth of hardened layer by flame hardening of steel wheel?

- (1) 0.15 mm
- (2) 0.8 mm
- (3) 0.35 mm
- (4) 1.5 mm

147. By what process maximum hardness is obtained for a steel part?

- (1) Carburizing
- (2) Nitriding
- (3) Cyaniding
- (4) Same hardness in all processes

148. The ratio of E of a wheel of mild steel to E of a bulk material is

- (1) 4
- (2) 2
- (3) 40
- (4) 50



149. Which of the following is a possible reason for flushing an A/C condenser?

- (1) To replace the old refrigerant with new refrigerant
- (2) To remove sludge and debris
- (3) To remove moisture
- (4) To improve cooling efficiency



150. The disadvantage of radial type compared to cross type is

- (1) uncomfortable ride at low speeds
- (2) uncomfortable ride at high speeds
- (3) higher cornering power
- (4) lower rolling resistance

